

# Coupled Hamiltonian Shear Degree Matrices in $\mathbb{A}^4$ : An Asymmetric $g \geq 5$ Family

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## Abstract

We prove an exact degree formula for a fixed family of canonical polynomial automorphisms of four-dimensional affine space. Let  $K$  be algebraically closed of characteristic zero, let  $g \geq 5$ , and compose the two triangular Hamiltonian shears generated by  $V(q) = q_1^2 q_2^2 + q_1^g$  and  $W(p) = p_1^2 p_2^2 + p_2^g$ . The mixed supports force a genuinely coupled two-phase degree calculation. We show that the first shear selects the degree-transfer matrix  $A_g = ((g-1, 0), (2, 1))$ , the second selects  $B_g = ((1, 2), (0, g-1))$ , and the complete step is governed by  $C_g = B_g A_g$ , not by the off-diagonal matrix that records the two half-steps. A half-open ratio cone is invariant under  $C_g$ ; strict old-term gaps and positive coefficients in characteristic zero prevent cancellation at every iterate. The second  $q$ -coordinate sees the Perron class, so

$$\deg(F_g^n) = e_2^T C_g^n(1, 1)^T, \quad \lambda_1(F_g) = \rho(C_g) = (\sqrt{g} + 1)^2.$$

Each elementary shear has degree  $g-1$ , whereas  $(\sqrt{g} + 1)^2 < (g-1)^2$ . The phrase *non-product* is used only for the displayed coordinate support and this strict degree comparison; no universal non-conjugacy statement is claimed.

## 1 Introduction

Polynomial automorphisms admit degree-growth mechanisms that are invisible if one only records the degree of each factor. In the planar generalized Hénon setting, reduced-word structure leads to a degree product law; this is a useful baseline, not a theorem about every higher-dimensional composition (Friedland and Milnor, 1989; Arnold Mathematical Journal, 2024). In higher dimension, several growth regimes occur and algebraic stability requires hypotheses that must be checked rather than inferred from a formal matrix (Déserti, 2018; Guedj and Sibony, 2002; Favre and Jonsson, 2011; Dang and Favre, 2021). Our aim is to isolate one small family in which all of the checks can be made explicitly.

We study a composition of two canonical shears on  $\mathbb{A}_K^4$ , with  $q = (q_1, q_2)$  and  $p = (p_1, p_2)$ . The potentials

$$V(q) = q_1^2 q_2^2 + q_1^g, \quad W(p) = p_1^2 p_2^2 + p_2^g$$

are asymmetric: one pure power occurs in the first  $q$ -gradient row and the other in the second  $p$ -gradient row. The mixed quartic monomials keep the two coordinates coupled. We fix the word  $F_g = T \circ S$  before doing any degree analysis:

$$S(q, p) = (q, p + \nabla V(q)), \quad T(q, p) = (q + \nabla W(p), p).$$

This order matters. The first shear produces an intermediate  $p$ -degree vector, and the second shear consumes that vector before the complete iterate returns to a  $q$ -degree vector.

The central point is a phase-sensitive selector calculation. If  $u = (u_1, u_2)^\top$  records the degrees of the  $q$ -coordinates, the selected rows of the first shear are

$$A_g = \begin{pmatrix} g-1 & 0 \\ 2 & 1 \end{pmatrix}.$$

After the intermediate vector  $v = A_g u$  is formed, the second shear selects

$$B_g = \begin{pmatrix} 1 & 2 \\ 0 & g-1 \end{pmatrix}.$$

The full return map is therefore

$$C_g = B_g A_g = \begin{pmatrix} g+3 & 2 \\ 2(g-1) & g-1 \end{pmatrix}.$$

The matrix  $C_g$ , rather than a half-step matrix, is the object whose Perron root can control the degree sequence.

Three technical issues make the statement nonformal. First, the selected faces must dominate both competing monomials and the carried coordinate. This requires a cone in the  $q$ -phase and a separate comparison in the intermediate  $p$ -phase. Second, the cone must be invariant under the complete matrix for every  $g \geq 5$ , not just at one initial step. Third, a spectral radius is not automatically a total degree: one must prove that the initial vector reaches the positive Perron class and that a concrete coordinate functional sees it. We address these points by strict inequalities and an explicit visibility argument.

The paper makes four bounded, internally provable contributions:

1. It gives an explicit symplectic polynomial automorphism with triangular inverses and a connected mixed support in the displayed split.
2. It proves a two-phase Newton-face selector for every integer  $g \geq 5$ , including the carried-coordinate inequalities and an invariant ratio cone.
3. It proves coefficientwise no-cancellation and the exact complete-step recurrence  $u_{n+1} = C_g u_n$ , then identifies total degree through the  $q_2$ -coordinate.
4. It obtains the closed form  $\lambda_1(F_g) = (\sqrt{g} + 1)^2$  and compares it strictly with  $\deg(S) \deg(T) = (g-1)^2$ , without making a universal product or priority claim.

The nearest four-dimensional coupled-symplectic literature studies a different question, namely horseshoes and uniform hyperbolicity of coupled Hénon maps (Fujioka et al., 2023). We study algebraic degree and prove the relevant formula from the displayed supports. The remainder of the paper is organized around that proof: Section 2 fixes the bounded comparison boundary; Section 3 defines the family and phase notation; Sections 5–6 establish canonicity and face selection; Sections 7–8 prove the recurrence and its observable degree; and Sections 10–12 state the narrow non-product comparison, limitations, and conclusion.

## 2 Bounded related work and collision boundary

**Planar degree products.** The plane classification of polynomial automorphisms provides the classical degree-product intuition for reduced compositions (Friedland and Milnor, 1989). We use

that result only as a two-dimensional baseline. Our map lives in dimension four, is constrained to be canonical, and has a mixed support that produces a non-diagonal complete-step matrix. We do not infer a contradiction with the planar law and do not claim a classification of higher-dimensional degree growth.

**Higher-dimensional degree growth.** Examples in dimension at least three show that degree sequences need not fall into the simplest planar regimes (Déserti, 2018). General terminology for algebraic stability and projective degree growth is useful here, but it does not replace the face-by-face induction needed for this family (Guedj and Sibony, 2002). Valuation and compactification methods in dimension two provide historical context for degree growth (Favre and Jonsson, 2011); they are not invoked in the proof below.

**Spectral language.** Dynamical degrees are often interpreted through spectral radii under explicit stability or visibility hypotheses (Guedj and Sibony, 2002; Dang and Favre, 2021). In our setting the relevant operator is finite and elementary, but the same logical caution applies: the positive matrix alone is insufficient. We prove that its Perron class is reached from  $(1, 1)^T$ , and that  $e_2^T$  observes the maximal coordinate degree.

**A four-dimensional neighbor.** A coupled-Hénon literature in four dimensions studies symplectic coupling for topological horseshoes and uniform hyperbolicity (Fujioka et al., 2023). This is a neighboring object, not a collision: our potentials, degree-transfer matrices, cone, and closed-form algebraic degree are different, and no hyperbolicity or topological-entropy statement appears here.

## 2.1 Paper12–Paper19 boundary

The following table records the exact object boundary used in the bounded collision review. Each row is empty at that scope; the table is not an exhaustive priority search.

| Ledger | Neighboring object                                                                  | Paper20 boundary                                                              |
|--------|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| P12    | Residue/period-three arithmetic and clock strata                                    | No residue classes, period-three data, or arithmetic clocks.                  |
| P13    | Primitive exact-period covers and cyclic orbit quotients                            | No periodic points, orbit quotients, or dynatomic schemes.                    |
| P14    | Support-one torus escape and finite-rank map geometry                               | No torus survivors, character lattices, or escape classification.             |
| P15    | Pure-trace/Jacobian fibers and quasi-finite loci                                    | No trace coordinates, Jacobian fibers, or ramification.                       |
| P16    | Support-size torus escape bounds                                                    | No support-size counting or finite-rank torus map.                            |
| P17    | Fixed-lag shift-like recurrence and torus-coset dimension                           | No recurrence variety, lag word, character extinction, or coset dimension.    |
| P18    | Marked trace coordinates and scheme-theoretic boundary ramification                 | No trace marking, boundary scheme, or branch divisor.                         |
| P19    | Maximum-dimensional translates, coefficientwise moduli, support-one GCD obstruction | No translates, GCD obstruction, effective height, or periodic classification. |

The paper also excludes the prime-clock, cat-map, centralizer, zeta, and quantization lines from Papers1–11 by scope. If a later review identifies a neighboring result, the response is to narrow the claim and add a boundary, not to erase the comparison.

### 3 Fixed family, coordinates, and phase notation

Throughout,  $K$  is algebraically closed and has characteristic zero, and  $g \geq 5$  is an integer. We work on  $\mathbb{A}_K^4$  with coordinates  $(q_1, q_2, p_1, p_2)$  and standard symplectic form

$$\omega = dq_1 \wedge dp_1 + dq_2 \wedge dp_2.$$

Set

$$V(q) = q_1^2 q_2^2 + q_1^g, \quad W(p) = p_1^2 p_2^2 + p_2^g.$$

The two triangular maps and their composition are

$$S(q, p) = (q, p + \nabla V(q)), \quad T(q, p) = (q + \nabla W(p), p), \quad F_g = T \circ S.$$

Writing  $\hat{p} = p + \nabla V(q)$ , the coordinates of one complete step are

$$\hat{p}_1 = p_1 + 2q_1 q_2^2 + gq_1^{g-1}, \quad \hat{p}_2 = p_2 + 2q_1^2 q_2, \quad (3.1)$$

$$F_{g,1} = q_1 + 2\hat{p}_1 \hat{p}_2^2, \quad F_{g,2} = q_2 + 2\hat{p}_1^2 \hat{p}_2 + g\hat{p}_2^{g-1}, \quad (3.2)$$

$$F_{g,3} = \hat{p}_1, \quad F_{g,4} = \hat{p}_2. \quad (3.3)$$

The formulas, rather than a formal Jacobian, determine the degree-transfer rows below.

For a polynomial  $h$ ,  $\deg h$  means total degree. For a polynomial map,  $\deg F$  is the maximum of its coordinate degrees. Define

$$q_{i,n} = q_i \circ F_g^n, \quad p_{i,n} = p_i \circ F_g^n,$$

and set

$$u_n = \begin{pmatrix} \deg q_{1,n} \\ \deg q_{2,n} \end{pmatrix}.$$

The intermediate vector  $v_{n+1}$  records the  $p$ -degrees immediately after the  $S$ -phase applied to  $F_g^n$ , before the  $T$ -phase:

$$v_{n+1} = \begin{pmatrix} \deg(p_{1,n} + \partial_{q_1} V(q_n)) \\ \deg(p_{2,n} + \partial_{q_2} V(q_n)) \end{pmatrix}.$$

For the carried initial coordinates we set  $u_0 = v_0 = (1, 1)^\top$ . Thus  $v_n$  and  $u_n$  have different phase meanings even when their entries happen to have comparable size.

The four selected monomials and their degree rows are

| phase row | selected monomial | degree in the input vector |
|-----------|-------------------|----------------------------|
| $S, 1$    | $q_1^g$           | $(g-1)u_1$                 |
| $S, 2$    | $q_1^2 q_2^2$     | $2u_1 + u_2$               |
| $T, 1$    | $p_1^2 p_2^2$     | $v_1 + 2v_2$               |
| $T, 2$    | $p_2^g$           | $(g-1)v_2$                 |

Consequently,

$$A_g = \begin{pmatrix} g-1 & 0 \\ 2 & 1 \end{pmatrix}, \quad B_g = \begin{pmatrix} 1 & 2 \\ 0 & g-1 \end{pmatrix}.$$

These rows are Newton-face exponent rows for the displayed monomials. They are not free Jacobian data, and every use of them below is accompanied by a strict selector inequality.

**Definition 3.1** (Two-phase cone). For  $u = (u_1, u_2)^\top \in \mathbb{R}_{>0}^2$ , write  $r(u) = u_2/u_1$  and define

$$\mathcal{C}_g = \left\{ u \in \mathbb{R}_{>0}^2 : 1 \leq r(u) < \frac{g-2}{2} \right\}.$$

The  $q$ -phase is the cone  $\mathcal{C}_g$ ; the intermediate  $p$ -phase is the image  $A_g \mathcal{C}_g$ , with its own selector inequalities.

**Theorem 3.2** (Asymmetric coupled-shear degree matrix). For every integer  $g \geq 5$ ,  $F_g$  is a symplectic polynomial automorphism of  $\mathbb{A}_K^4$ . If

$$C_g = B_g A_g = \begin{pmatrix} g+3 & 2 \\ 2(g-1) & g-1 \end{pmatrix},$$

then, for every  $n \geq 1$ ,

$$\deg(F_g^n) = e_2^\top C_g^n \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad (3.4)$$

and

$$\lambda_1(F_g) = \lim_{n \rightarrow \infty} \deg(F_g^n)^{1/n} = \rho(C_g) = (\sqrt{g} + 1)^2. \quad (3.5)$$

Each elementary shear has degree  $g-1$ , so

$$\lambda_1(F_g) < (g-1)^2 = \deg(S) \deg(T).$$

The mixed support makes the displayed coordinate split coupled. No non-conjugacy statement is asserted for other coordinate systems.

## 4 Degree bookkeeping and support filtration

We use ordinary total degree, but it is helpful to separate the elementary properties of degree that make the selector argument rigorous. If  $u = (u_1, u_2) \in \mathbb{R}_{\geq 0}^2$  and  $m = q_1^{a_1} q_2^{a_2}$ , define its  $u$ -weight by

$$\text{wt}_u(m) = a_1 u_1 + a_2 u_2.$$

For a polynomial in  $q_1, q_2$ , its  $u$ -weighted degree is the maximum of the weights of its monomials with nonzero coefficient. Substitution of coordinate polynomials of degrees  $u_1, u_2$  gives an upper bound by this weighted degree. If a monomial attaining the maximum has a nonzero coefficient and no competing summand has the same degree with an opposite coefficient, the upper bound is attained.

The only weighted-degree comparisons used in this paper are finite. For the first gradient row they are

$$\text{wt}_u(q_1 q_2^2) = u_1 + 2u_2, \quad \text{wt}_u(q_1^{g-1}) = (g-1)u_1.$$

For the second shear they are

$$\text{wt}_v(p_1 p_2^2) = v_1 + 2v_2, \quad \text{wt}_v(p_1^2 p_2) = 2v_1 + v_2, \quad \text{wt}_v(p_2^{g-1}) = (g-1)v_2.$$

The strict inequalities in Sections 6 and 7 say that the displayed maximum is unique at the level of degree. They are stronger than an assertion that a row is generically selected: they hold for every degree vector in the invariant cone.

**Lemma 4.1** (Max-plus substitution bound). *Let  $h = \sum_{\alpha} c_{\alpha} x^{\alpha}$  and let  $x_i$  be replaced by polynomials of degrees  $d_i$ . Then*

$$\deg h \leq \max_{\alpha: c_{\alpha} \neq 0} \sum_i \alpha_i d_i.$$

*If one monomial on the right has a coefficient that remains nonzero and has strictly larger degree than all other summands in the same addition, equality holds.*

*Proof.* The degree of a product is the sum of the degrees of its factors, and the degree of a sum is at most the maximum of the summand degrees. Applying these facts to each monomial and then taking the maximum proves the bound. A strict maximum cannot be cancelled by a lower-degree summand; if its coefficient is nonzero, it realizes the bound.  $\square$

*Remark 4.2* (Why a face row is not a Jacobian row). The matrices  $A_g$  and  $B_g$  encode weighted degrees of four specific support monomials after differentiating the potentials. A Jacobian matrix records first derivatives of the full map and has a different meaning. In particular, replacing  $A_g$  or  $B_g$  by a Jacobian or Hessian matrix would discard the carried-coordinate comparisons and would not prove (6.6).

## 4.1 The phase filtration

The complete step has the filtration

$$\underbrace{u_n}_{q\text{-phase}} \xrightarrow{S} \underbrace{v_{n+1}}_{p\text{-phase}} \xrightarrow{T} \underbrace{u_{n+1}}_{q\text{-phase}}.$$

The first arrow is not an iteration of  $F_g$ , and the second arrow cannot be applied to  $u_n$  directly. This distinction is relevant even if one only wants the maximal degree: the inequalities for the  $S$ -phase involve the ratio  $u_2/u_1$ , whereas the second  $T$ -row involves  $v_2/v_1 = (2 + u_2/u_1)/(g - 1)$ . The latter ratio is not the former ratio.

The initial values are deliberately assigned to both phase ledgers:  $u_0 = v_0 = (1, 1)^{\top}$ . The value  $v_0$  represents the original degree-one  $p$ -coordinates only for the base comparison. For every genuine step  $n \rightarrow n + 1$ ,  $v_{n+1}$  is defined by the  $S$ -phase and is then consumed by  $T$ . This convention lets the base case and the induction use the same displayed inequalities without silently identifying the phases.

## 4.2 A useful matrix comparison

The complete matrix is positive, but the phase matrices are only nonnegative:

$$A_g = \begin{pmatrix} g-1 & 0 \\ 2 & 1 \end{pmatrix}, \quad B_g = \begin{pmatrix} 1 & 2 \\ 0 & g-1 \end{pmatrix}.$$

The off-diagonal stacked matrix

$$M_g = \begin{pmatrix} 0 & B_g \\ A_g & 0 \end{pmatrix}$$

satisfies

$$M_g^2 = \begin{pmatrix} B_g A_g & 0 \\ 0 & A_g B_g \end{pmatrix}.$$

Thus  $\rho(M_g)^2 = \rho(B_g A_g)$ . A statement about one application of  $M_g$  would therefore describe a half-step; the degree per complete iterate is controlled by  $\rho(C_g)$ , not by  $\rho(M_g)$  without the square. This elementary observation prevents a common phase-counting error.

## 5 Canonical shears and explicit inverses

The canonicity calculation is short, but it fixes the algebraic object before the degree proof starts.

**Lemma 5.1** (Triangular inverses). *The maps  $S$  and  $T$  are polynomial automorphisms with*

$$S^{-1}(q, p) = (q, p - \nabla V(q)), \quad T^{-1}(q, p) = (q - \nabla W(p), p).$$

*Consequently  $F_g^{-1} = S^{-1} \circ T^{-1}$  is polynomial.*

*Proof.* The  $q$ -coordinates are unchanged by  $S$ , so subtracting the same gradient recovers  $p$ . The  $p$ -coordinates are unchanged by  $T$ , so subtracting  $\nabla W(p)$  recovers  $q$ . Composition gives the displayed inverse of  $F_g$ .  $\square$

**Lemma 5.2** (Symplecticity). *Both  $S$  and  $T$ , hence  $F_g$ , preserve  $\omega$ .*

*Proof.* For  $S$ ,

$$\begin{aligned} S^* \omega &= \sum_{i=1}^2 dq_i \wedge d(p_i + \partial_{q_i} V(q)) \\ &= \omega + \sum_{i,j=1}^2 \partial_{q_i q_j} V(q) dq_i \wedge dq_j. \end{aligned}$$

The Hessian of  $V$  is symmetric, while  $dq_i \wedge dq_j$  is antisymmetric, so the last sum is zero. The same calculation, with  $q, p$  interchanged and  $W$  in place of  $V$ , gives  $T^* \omega = \omega$ . Therefore  $F_g^* \omega = S^*(T^* \omega) = \omega$ .  $\square$

*Remark 5.3.* The characteristic-zero assumption is not needed for the formal Hessian-symmetry identity. It enters in Lemma 8.1, where the positive integer coefficients 2 and  $g$  must remain nonzero.

## 6 The two Newton-face selectors

We now prove that the rows  $A_g$  and  $B_g$  are forced by the displayed supports on the relevant cone. The argument is degree-theoretic: a selected monomial supplies a lower bound, and the strict inequalities bound every competing summand from above.

### 6.1 The $S$ -phase

**Lemma 6.1** (First-shear selector). *If  $u \in \mathcal{C}_g$ , then the gradient contribution in the  $S$ -phase has degree vector  $A_g u$ . Moreover, if the carried intermediate vector is the one from the preceding step, the selected gradient strictly dominates that carried vector.*

*Proof.* The first gradient coordinate is

$$\partial_{q_1} V = 2q_1 q_2^2 + g q_1^{g-1}.$$

Its two candidate degrees are  $u_1 + 2u_2$  and  $(g-1)u_1$ . Since  $u_2/u_1 < (g-2)/2$ ,

$$u_1 + 2u_2 < (g-1)u_1. \tag{6.1}$$

Thus  $q_1^g$  selects the first row. The second gradient coordinate is

$$\partial_{q_2} V = 2q_1^2 q_2,$$

whose degree is  $2u_1 + u_2$ , giving  $A_g u$ .

It remains to include the carried  $p$ -coordinates. At the initial step, the carried degrees are one and

$$(g-1)u_{0,1} = g-1 > 1, \quad 2u_{0,1} + u_{0,2} = 3 > 1.$$

For a later step, the preceding intermediate vector is  $v_n = A_g u_{n-1}$ . Lemma 7.1 gives  $u_n > u_{n-1}$  componentwise. Since  $A_g$  has nonnegative entries,  $A_g u_n > A_g u_{n-1} = v_n$ . Thus the selected gradient dominates both carried coordinates strictly.  $\square$

## 6.2 The $T$ -phase

**Lemma 6.2** (Second-shear selector). *If  $u \in \mathcal{C}_g$  and  $v = A_g u$ , then the gradient contribution in the  $T$ -phase has degree vector  $B_g v$ , and it strictly dominates the carried  $q$ -coordinates.*

*Proof.* The first  $p$ -gradient coordinate is

$$\partial_{p_1} W = 2p_1 p_2^2,$$

so its degree is  $v_1 + 2v_2$ . It dominates the carried  $q_1$ -degree because

$$v_1 + 2v_2 - u_1 = (g-1)u_1 + 2(2u_1 + u_2) - u_1 = (g+2)u_1 + 2u_2 > 0. \quad (6.2)$$

For the second row,

$$\partial_{p_2} W = 2p_1^2 p_2 + g p_2^{g-1}.$$

The cross term has degree  $2v_1 + v_2$ , while the pure term has degree  $(g-1)v_2$ . If  $r = u_2/u_1$ , then

$$v_1 = (g-1)u_1, \quad v_2 = (2+r)u_1,$$

and therefore

$$\frac{v_2}{v_1} = \frac{2+r}{g-1} \geq \frac{3}{g-1} > \frac{2}{g-2},$$

where the final inequality is equivalent to  $g > 4$ . Hence

$$2v_1 + v_2 < (g-1)v_2. \quad (6.3)$$

The pure term selects the second row. It also dominates the carried  $q_2$ -degree because

$$(g-1)v_2 - u_2 = 2(g-1)u_1 + (g-2)u_2 > 0. \quad (6.4)$$

$\square$

## 6.3 The complete matrix and the half-step warning

Combining the selected rows gives

$$B_g A_g = \begin{pmatrix} 1 & 2 \\ 0 & g-1 \end{pmatrix} \begin{pmatrix} g-1 & 0 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} g+3 & 2 \\ 2(g-1) & g-1 \end{pmatrix} = C_g. \quad (6.5)$$

The phase recurrence is

$$v_{n+1} = A_g u_n, \quad u_{n+1} = B_g v_{n+1} = C_g u_n. \quad (6.6)$$

One can instead stack the two phase vectors and write a formal off-diagonal matrix, but that matrix advances only one shear at a time. Its nonzero spectrum must be interpreted with a two-phase period correction. Equation (6.5) is the complete-step theorem matrix.

## 7 Two-stage cone invariance

The selectors above are conditional on a cone. We now show that the complete matrix returns the  $q$ -degree vector to that same cone for every iterate.

**Lemma 7.1** (Ratio map). *Let*

$$C_g = \begin{pmatrix} g+3 & 2 \\ 2(g-1) & g-1 \end{pmatrix}.$$

*If  $u \in \mathcal{C}_g$ , then  $C_g u \in \mathcal{C}_g$  and  $C_g u > u$  componentwise.*

*Proof.* Write  $r = u_2/u_1$  and  $R = (g-2)/2$ . The new ratio is

$$f_g(r) = \frac{(C_g u)_2}{(C_g u)_1} = \frac{(g-1)(2+r)}{g+3+2r}.$$

Its derivative is

$$f'_g(r) = \frac{(g-1)^2}{(g+3+2r)^2} > 0.$$

At the lower endpoint,

$$f_g(1) = \frac{3(g-1)}{g+5} > 1$$

for  $g > 4$ . At the upper endpoint,

$$f_g(R) = \frac{(g-1)(g+2)}{2(2g+1)}.$$

After multiplying by the positive denominator, the inequality  $f_g(R) < R$  is equivalent to

$$(g-1)(g+2) < (g-2)(2g+1),$$

whose right-minus-left side is  $g(g-4) > 0$ . Monotonicity therefore gives  $1 \leq f_g(r) < R$  whenever  $1 \leq r < R$ . Finally,

$$(C_g u)_1 - u_1 = (g+2)u_1 + 2u_2 > 0,$$

$$(C_g u)_2 - u_2 = 2(g-1)u_1 + (g-2)u_2 > 0.$$

This proves cone invariance and strict componentwise growth.  $\square$

*Remark 7.2* (Why the cone has two phases). It is tempting to write one strict cone in which both  $A_g u > v$  and  $B_g v > u$  hold. That is not the phase structure of the map. The intermediate vector is exactly  $v_{n+1} = A_g u_n$ ; it is not a variable that should be compared strictly with itself. The correct induction has a  $q$ -phase cone, followed by the  $S$ -selector, followed by the  $T$ -selector, and then a complete-step return through  $C_g$ .

### 7.1 Old-term dominance

The strict face gaps also control all carried coordinates. We record them together because this is where a formal degree matrix could otherwise lose an old term.

**Proposition 7.3** (All carried terms are dominated). *For every  $n \geq 0$ , the following inequalities hold:*

$$\max\{v_{1,n}, u_{1,n} + 2u_{2,n}\} < (g-1)u_{1,n}, \quad v_{2,n} < 2u_{1,n} + u_{2,n}, \quad (7.1)$$

$$u_{1,n} < v_{1,n+1} + 2v_{2,n+1}, \quad \max\{u_{2,n}, 2v_{1,n+1} + v_{2,n+1}\} < (g-1)v_{2,n+1}. \quad (7.2)$$

Here  $u_0 = v_0 = (1, 1)^\top$ , and  $v_{n+1} = A_g u_n$ .

*Proof.* The inequalities involving a new gradient term are (6.1), (6.2), (6.3), and (6.4). For the carried terms at the initial step, the degrees are one, while  $g-1 > 1$  and  $3 > 1$ . At a later step,  $v_n = A_g u_{n-1}$ , and Lemma 7.1 gives  $u_n > u_{n-1}$  componentwise. Applying the nonnegative matrix  $A_g$  gives  $v_n < A_g u_n$ , which proves both carried inequalities in the  $S$ -phase. The carried  $q$ -terms in the  $T$ -phase are exactly the two gaps (6.2) and (6.4). The cross term in the second row is strictly below the pure term by (6.3). This closes the simultaneous induction.  $\square$

## 8 No cancellation, exact recurrence, and visibility

### 8.1 Coefficientwise no-cancellation

**Lemma 8.1** (Positive coefficients prevent cancellation). *For every  $n$ , every coordinate polynomial of  $F_g^n$  has coefficients in  $\mathbb{Z}_{\geq 0}$ . Every monomial selected in (6.6) has a positive coefficient, and its degree cannot be removed by cancellation.*

*Proof.* The initial coordinates have coefficient one. The two shears use only addition, multiplication, and multiplication by the positive integers 2 and  $g$ . Induction therefore keeps all coefficients in  $\mathbb{Z}_{\geq 0}$ . Proposition 7.3 gives a strict degree gap between each selected monomial and every competing summand. A selected monomial appears with a positive coefficient in the corresponding gradient summand, so its coefficient in the sum remains positive. Since  $\text{char } K = 0$ , no positive integer coefficient becomes zero after passing to  $K$ . The argument needs positivity and a strict gap; it does not assert that two distinct expansion paths always yield distinct monomials.  $\square$

### 8.2 The exact complete-step recurrence

**Proposition 8.2** (Two-phase recurrence). *For every  $n \geq 0$ ,*

$$v_{n+1} = A_g u_n, \quad u_{n+1} = C_g u_n.$$

Consequently  $u_n = C_g^n (1, 1)^\top$ .

*Proof.* Lemma 6.1 identifies the exact  $S$ -phase vector. Lemma 6.2 identifies the exact  $T$ -phase vector once  $v_{n+1} = A_g u_n$  is known. Lemma 7.1 keeps  $u_n$  in the selector cone, while Proposition 7.3 and Lemma 8.1 ensure that no carried term or cancellation changes a selected degree. Equation (6.6) follows, and induction from  $u_0 = (1, 1)^\top$  gives the closed recurrence.  $\square$

**Lemma 8.3** (Visibility of the total degree). *For every  $n \geq 1$ , the second  $q$ -coordinate has maximal degree among all four coordinates of  $F_g^n$ . In particular,*

$$\deg(F_g^n) = e_2^\top u_n.$$

*Proof.* Cone invariance gives  $u_{n,2} > u_{n,1}$  for  $n \geq 1$ . The final  $p$ -degrees are  $v_n = A_g u_{n-1}$ . Direct subtraction gives

$$C_g - A_g = \begin{pmatrix} 4 & 2 \\ 2g-4 & g-2 \end{pmatrix},$$

whose entries are positive for  $g \geq 5$ . Therefore  $u_n = C_g u_{n-1} > A_g u_{n-1} = v_n$  componentwise. Both final  $p$ -coordinates are below the corresponding  $q$ -coordinates, and the second  $q$ -degree is larger than the first. The claim follows.  $\square$

## 9 Perron calculation and dynamical degree

The recurrence is now exact, so the asymptotic calculation is a finite matrix computation with an explicitly visible coordinate.

**Lemma 9.1** (Perron root). *The eigenvalues of  $C_g$  are*

$$\lambda_{\pm} = g + 1 \pm 2\sqrt{g} = (\sqrt{g} \pm 1)^2.$$

*The larger root is the Perron root.*

*Proof.* The trace and determinant are

$$\text{tr}(C_g) = 2g + 2, \quad \det(C_g) = (g + 3)(g - 1) - 4(g - 1) = (g - 1)^2.$$

Thus the characteristic discriminant is

$$(2g + 2)^2 - 4(g - 1)^2 = 16g,$$

which gives the displayed roots. All four entries of  $C_g$  are positive, so the larger eigenvalue is simple and has a positive eigenvector by the Perron–Frobenius theorem.  $\square$

*Proof of Theorem 3.2.* Lemmas 5.1 and 5.2 give polynomial invertibility and preservation of  $\omega$ . Lemmas 6.1–8.1 give the exact complete-step recurrence, and Proposition 8.2 gives  $u_n = C_g^n(1, 1)^T$ . Lemma 8.3 identifies the total degree with the second component, proving (3.4). Lemma 9.1 supplies the positive root. Because both the initial vector and  $e_2$  have positive pairing with the Perron class, the subdominant eigenvalue is exponentially negligible and

$$\lim_{n \rightarrow \infty} \left( e_2^T C_g^n(1, 1)^T \right)^{1/n} = \rho(C_g).$$

Finally, the pure powers  $q_1^{g-1}$  in  $\nabla V$  and  $p_2^{g-1}$  in  $\nabla W$  show  $\deg(S) = \deg(T) = g - 1$ . The strict comparison is

$$(g - 1)^2 - (\sqrt{g} + 1)^2 = \sqrt{g}(\sqrt{g} - 2)(\sqrt{g} + 1)^2 > 0$$

for  $g \geq 5$ . The mixed monomials join the two coordinate labels in the displayed support, which is the stated narrow coupling claim.  $\square$

## 9.1 A scalar recurrence and an exact closed form

The matrix formula also gives a useful scalar check on the degree sequence. Set

$$d_n = e_2^T C_g^n \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \deg(F_g^n) \quad (n \geq 1),$$

and extend the notation by  $d_0 = 1$ .

**Proposition 9.2** (Scalar degree recurrence). *The sequence  $(d_n)$  satisfies*

$$d_{n+2} = 2(g+1)d_{n+1} - (g-1)^2 d_n, \quad d_0 = 1, \quad d_1 = 3(g-1).$$

*Equivalently, with  $\lambda_{\pm} = (\sqrt{g} \pm 1)^2$ ,*

$$d_n = \frac{3(g-1) - \lambda_-}{\lambda_+ - \lambda_-} \lambda_+^n + \frac{\lambda_+ - 3(g-1)}{\lambda_+ - \lambda_-} \lambda_-^n.$$

*Proof.* The Cayley–Hamilton identity for  $C_g$  is

$$C_g^2 - 2(g+1)C_g + (g-1)^2 I = 0.$$

Multiplying on the left by  $e_2^T C_g^n$  and on the right by  $(1, 1)^T$  gives the recurrence. Since

$$C_g \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} g+5 \\ 3(g-1) \end{pmatrix},$$

the initial values follow. Solving the two-by-two linear system for the coefficients of  $\lambda_+^n$  and  $\lambda_-^n$  gives the formula.  $\square$

The closed form is not needed to establish the dynamical degree, but it is a useful internal consistency check: the coefficient of the Perron term is positive because  $C_g$  is positive and both  $e_2$  and the initial vector pair positively with its Perron eigenvectors. It also makes clear why the visible-coordinate argument in Lemma 8.3 is part of the theorem rather than an after-the-fact interpretation of a matrix sequence.

## 9.2 Why the threshold is $g \geq 5$

The lower threshold is forced by the second selector. The strict inequality

$$\frac{3}{g-1} > \frac{2}{g-2}$$

is equivalent to  $g > 4$ . At  $g = 4$ , the endpoint comparison becomes a tie for the smallest allowed ratio, and the  $p_2^g$  face is not separated from the mixed  $p_1^2 p_2$  face by the argument above. At the same time, the upper-endpoint cone calculation loses its strict margin because  $g(g-4) = 0$ . We therefore state the theorem only for  $g \geq 5$ ; no continuation to  $g = 4$  is hidden in the notation.

The characteristic-zero and coefficient hypotheses have the same status. The formal degree inequalities are combinatorial, but the passage from a selected summand to a nonzero leading coefficient uses the positive integers 2 and  $g$ . A sign-changing potential or a positive-characteristic specialization would require a new no-cancellation argument and is outside the present theorem.

## 10 The narrow non-product comparison

The phrase *non-product* has a deliberately restricted meaning. The support of  $V$  contains  $q_1^2 q_2^2$ , and the support of  $W$  contains  $p_1^2 p_2^2$ . Thus the support graph on the two labels in each phase is connected. At the degree level, this connectivity appears as the positive off-diagonal entries of  $C_g$ . If the phases were independent scalar recurrences in the displayed split, the complete transfer matrix would be diagonal after ordering the two labels; that is not what (6.5) gives.

The comparison with the elementary degree product is equally specific:

$$\lambda_1(F_g) = (\sqrt{g} + 1)^2 < (g - 1)^2 = \deg(S) \deg(T).$$

This inequality does not say that every coupled composition has submultiplicative degree growth, nor that  $F_g$  cannot be conjugated to a product in a different coordinate system. It records the exact value for this fixed word and the strict baseline comparison in the coordinates where the supports are given.

*Example 10.1* (The endpoint  $g = 5$ ). At the smallest allowed parameter,

$$A_5 = \begin{pmatrix} 4 & 0 \\ 2 & 1 \end{pmatrix}, \quad B_5 = \begin{pmatrix} 1 & 2 \\ 0 & 4 \end{pmatrix}, \quad C_5 = \begin{pmatrix} 8 & 2 \\ 8 & 4 \end{pmatrix}.$$

The eigenvalues are  $6 \pm 2\sqrt{5}$ , so

$$\lambda_1(F_5) = 6 + 2\sqrt{5} < 16 = (5 - 1)^2.$$

The cone is  $1 \leq r < 3/2$ ; the endpoint calculation in Lemma 7.1 is strict because  $5(5 - 4) > 0$ .

## 11 Limitations, anti-claims, and collision discipline

The proof is intentionally narrow. It does not cover arbitrary sparse potentials, arbitrary coefficients, arbitrary shear words, or positive characteristic. In particular, changing signs can invalidate the coefficientwise no-cancellation step, and changing the supports can invalidate the two selected rows or the ratio cone.

We do not claim a finite Newton-fan theorem for all symplectic polynomial automorphisms, a classification in dimension four, or equality between algebraic, topological, arithmetic, and measure-theoretic entropy. We do not study periodic points, traces, multipliers, invariant curves, centralizers, torus translates, arithmetic specializations, or effective orbit counts. The paper contains no numerical or CAS certificate.

The bounded P12–P19 table in Section 2 is a collision boundary, not an exhaustive priority search. The cited literature is context only: Friedland–Milnor gives the planar baseline; Déserti and Guedj–Sibony provide higher-dimensional context; Favre–Jonsson supplies historical valuation context; Dang–Favre motivates explicit spectral visibility; and Fujioka et al. provide a nearby four-dimensional hyperbolicity object. None of these sources proves the family, the selector inequalities, the no-cancellation statement, or the closed-form root.

The following conditions are hard stops for any revision:

1. a selector tie or an untracked carried coordinate at any iterate;
2. use of one strict cone across the phase-return equality  $v_{n+1} = A_g u_n$ ;
3. use of a half-step matrix without a period correction;

4. cancellation made possible by signs or positive characteristic;
5. failure to show that the Perron class is reachable and visible to total degree;
6. removal of the mixed support or introduction of a universal conjugacy claim;
7. promotion of a contextual citation to proof or priority evidence.

## 12 Conclusion

We fixed an explicit pair of Hamiltonian shears and proved its degree growth without appealing to a generic Newton-fan principle. The proof has four interlocking parts: symmetric Hessians give canonicity and triangular inverses; the literal support monomials give  $A_g$  and  $B_g$ ; a two-stage cone plus old-term dominance and positive coefficients closes the selector induction; and the complete positive matrix  $C_g$ , together with the visible  $q_2$ -coordinate, gives the Perron formula for the dynamical degree. The resulting value  $(\sqrt{g} + 1)^2$  is strictly below the product  $(g - 1)^2$  of the elementary shear degrees for every  $g \geq 5$ .

The conclusion is a theorem about this family and this coordinate-level comparison. Extending it to other supports or words would require a new selector and no-cancellation proof; it is not a consequence of the present argument.

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